PORTFOLIO PROJECT · ML PRODUCT · FULL STACK

Palate: a sommelier on the shelf

A wine recommendation SaaS that matches shoppers to bottles in stock at that store, right now

Dhruv Khanna · MS Business Analytics, Quinnipiac University

React · FastAPI · scikit-learn · Railway

THE PROBLEM

Shoppers can't tell what they'll like, and staff can't help everyone

−8%

U.S. wine volume in the 12 months to August 2024

35%

of non-drinkers can't tell which wines deliver which flavors

1

customer at a time is all a knowledgeable floor rep can serve

Sources: WSWA SipSource, Feb 2025 · Wine Market Council Benchmark Survey, Dec 2025

THE PRODUCT

Sixty seconds from question to bottle

Step 1

Taste quiz

Five questions: fruit, texture, sweetness, occasion, budget. No wine knowledge needed.

Step 2

Match

Scores every in-stock, in-budget wine against the shopper's answers.

Step 3

Top 3 picks

Each on the shelf today, each with a plain-English reason it was picked.

ARCHITECTURE

One app, one API, one model

Frontend

`React · Vite`

Quiz, results and a store dashboard for the owner.

API

`Python · FastAPI`

Endpoints for recommendations, inventory and analytics.

Model and data

`scikit-learn`

Wine profiles, live stock and every quiz session.

Backend deployed on Railway. A synthetic data generator creates realistic sessions for development and training before real stores come on board.

THE RECOMMENDATION ENGINE

It works on day one and learns from every sale

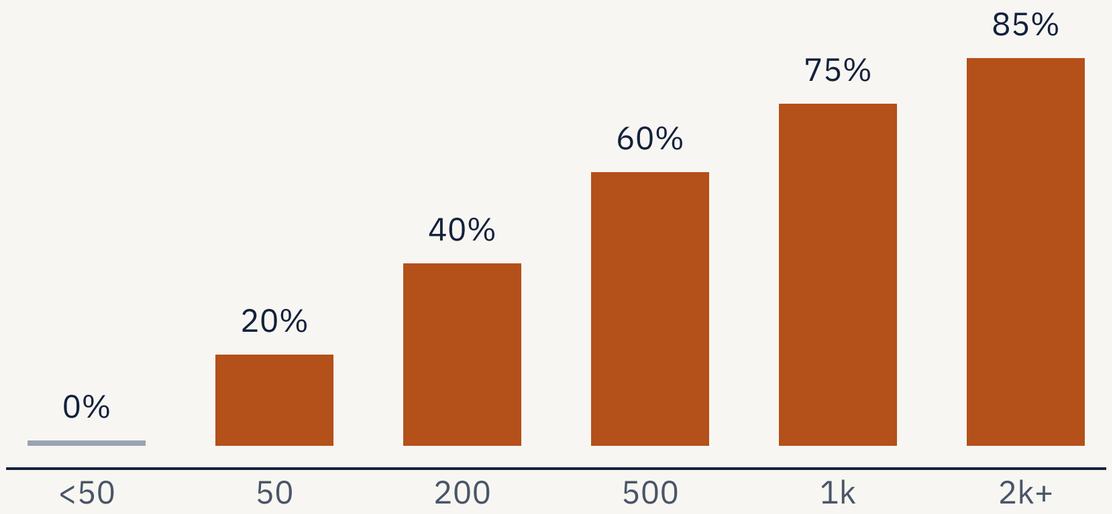
Stage 1 · Cosine similarity

Compares the shopper's answers with each wine's sensory profile. Needs no history and every score can be explained.

Stage 2 · K-nearest neighbors

Trained on quiz sessions that ended in a purchase. Eight features: five answers plus hour, weekday and quiz time.

Weight given to Stage 2



Purchased sessions at the store

DESIGN DECISIONS

Three choices that make it a retail product

Filter before scoring

Out-of-stock and over-budget wines are removed first. It never recommends what the store can't sell.

Explain every pick

Each wine comes with a one-line reason tied to the shopper's own answers.

One model per store

Each store's purchase history trains its own model, which competitors can't copy.

STATUS

Where it stands, and what comes next

Built

- Quiz, results and store dashboard screens
- Recommend, inventory and analytics APIs
- Two-stage scorer and KNN trainer
- Roadmap, tech spec and investor pitch

Next

- Pilot in one store to collect real purchases
- Measure how often a top-3 pick is the wine bought
- Compare Stage 1 alone against the blend